

Non

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Problem

Is a set of dyna

What is shadowing?

Theory of shadowing studies relations imate and exact trajectories of dynam unbounded time intervals. Shadowing sufficiently precise approximate trajec trajectory) is close to some exact traje

density of orbital s

y Osipov

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amical systems having son

between approx-
nical systems on
means that any
ctory (a pseudo-
ectory.

Previous results

	POTP
H(M)	generic Pilyugin, Plamenevskaya 1999
Diff(M)	nondense Bonatti, Diaz, Turcat 2000

Shadowing in C^1 -to

Chebyshev laboratory, Department
Saint Petersburg State University

the shadowing property generalizes

Case 1

Let F be a flow on a compact metric space X which is a flow shadowed by a flow G . Assume that G has the shadowing property. Let q_1 and q_2 be points in X .

OSP	WSP
generic Pilyugin, Plamenevskaya 1999	generic Pilyugin, Plamenevskaya 1999
?	generic Crovisier 2006

Topology.

Department of Mathematics and Mechanics,
University, Saint Petersburg, Russia

generic or nondense?

a diffeomorphism from $\mathcal{V} \subset \text{Diff}(S^2 \times S^1)$
assigned to some mild skew product G .

that there exist two hyperbolic periodic points
lying on different fibres such that

$$W^u(q_1) \cap W^s(q_2) \neq \emptyset.$$

POTP

Let f be a homeomorphism of a metric

Definition 1. A sequence $\{x_k\}_{k \in \mathbb{Z}}$ is a *d -pseudotrajectory* of f iff

$$\text{dist}(x_{k+1}, f(x_k)) < d, \quad \forall k$$

Definition 2. A diffeomorphism f has the *tracing property* or *standard shadowing* (POTP) iff for any $\epsilon > 0$ there exists a $\delta > 0$ such that for any δ -pseudotrajectory $\xi = \{x_k\}_{k \in \mathbb{Z}}$ and any point $p \in M$ such that

$$\text{dist}(x_k, f^k(p)) < \epsilon.$$

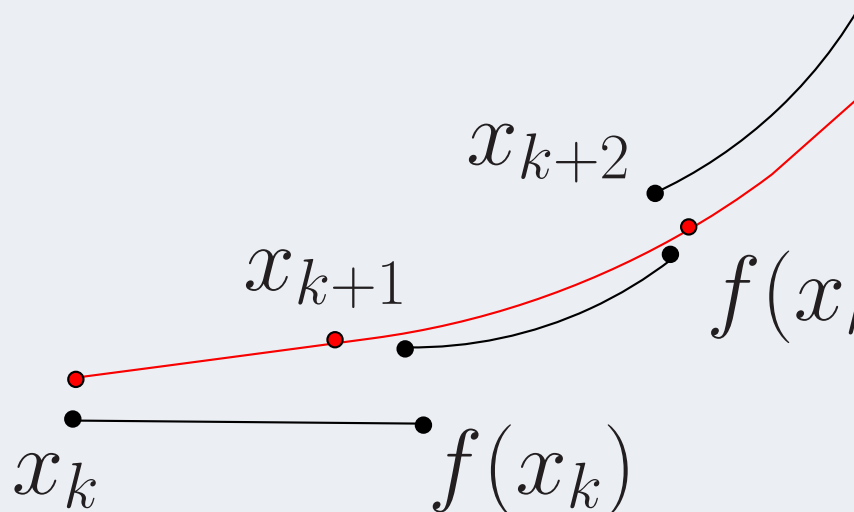


Figure: Example when POTP holds: black denotes a pseudotrajectory, red denotes an exact trajectory.

space (M, dist) .

of points of M is

$\in \mathbb{Z}$.

has *pseudorbit*

property ($f \in$

$d > 0$ such that

$\in \mathbb{Z}$ there exists a

$f(x_{k+2})$

$(k+1)$

notes a

ory.

Figure: Previous

Main result: C^1 -non

Theorem (Osipov, 20

$W \subset \text{Diff}(S^2 \times S^1)$ such
have OSP.

*It is possible to construct
for any n -dimensional M*

General strategy of

In essence, we use a
Gorodetski developed for
 $\text{Diff}(M)$ with specific pr

Step 1. Prove that th
neighborhood $\mathcal{U}$ in the
certain type such that a
not have OSP.

us results on the topic.

ndensity of OSP

010). *There exists a domain
ch that any $f \in W$ does not*

*ct similar domains in $\text{Diff}(M)$
manifold M with $n \geq 3$.*

the proof

strategy of Ilyashenko and
or construction of domains in
properties.

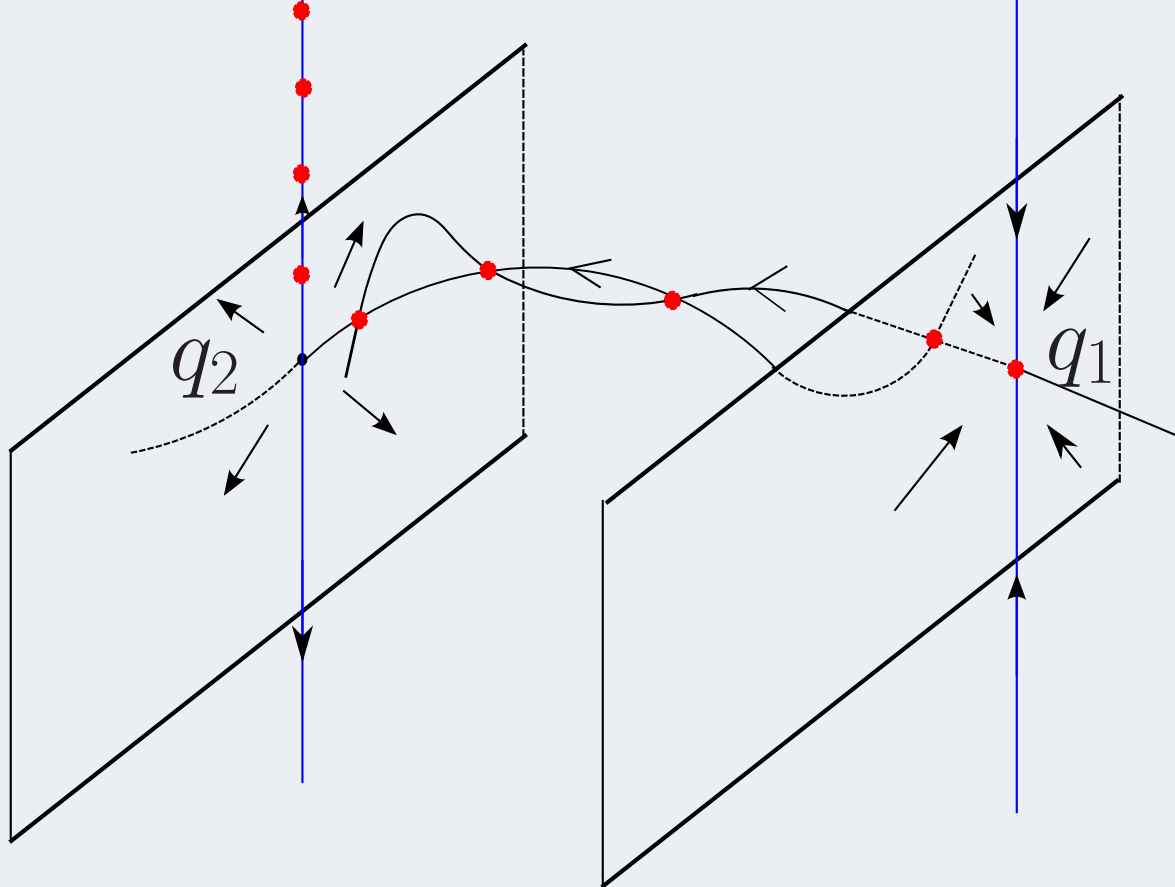
here exists a *sufficiently good*
space of skew products of a
any skew product from $\mathcal{U}$ does

Figure: F in
denotes fibre

Using expa
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not be orb
Note that
tant to pr
dotrajecto

Case 2

Let F be



Case 1: red denotes a pseudotrajectory, blue lines.

transitivity of the Bernoulli shift, we prove that the constructed pseudotrajectory (cf. the figure) can be made arbitrarily close to any exact trajectory.

In order to derive contradiction it is important to prove that the intermediate part of the pseudotrajectory is far from the points q_1 and q_2 .

a diffeomorphism from $\mathcal{V} \subset \text{Diff}(S^2 \times S^1)$

Any structurally stable diffeomorphism has a pseudo-trajectory, i.e. a sequence of points that approximates an exact trajectory.
(Robinson, Sawada, Morimoto, 1998)

OSP and WSP

By $N(\epsilon, A)$ denote the ϵ -neighborhood of A .
 $O(p, f) = \{f^k(p)\}_{k \in \mathbb{Z}}$ denote the trajectory of p under f .

Definition 3. A diffeomorphism f has the *Orbit Shadowing Property* ($f \in \text{OSP}$) iff for a given $\epsilon > 0$ there exists a $d > 0$ such that for any d -pseudo-orbit $\xi = \{x_k\}_{k \in \mathbb{Z}}$ there exists a point $p \in M$ such that

$$\xi \subset N(\epsilon, O(p, f)) \quad \text{and} \quad O(p, f) \subset N(\epsilon, \xi)$$

Example of $f \in \text{OSP} \setminus \text{POTP}$: irrational rotation on a circle.

Definition 4. A diffeomorphism f has the *Weak Shadowing Property* ($f \in \text{WSP}$) iff for a given $\epsilon > 0$ there exists a $d > 0$ such that for any d -pseudo-orbit $\xi = \{x_k\}_{k \in \mathbb{Z}}$ there exists a point $p \in M$ such that

$$\xi \subset N(\epsilon, O(p, f))$$

ism has *POTP* (977–80).

d of $A \subset M$. By
jectory of p .

has *orbital shad-*
ny $\epsilon > 0$ there
pseudotrajectory
 M such that

$) \subset N(\epsilon, \xi)$.

al rotation of the

has *weak shad-*
ny $\epsilon > 0$ there
pseudotrajectory
 M such that

Step 2. For transfer f
morphisms apply the res
informally formulated as

Theorem (Gorodetskiy)
*ciently good neighborhoods
ucts of a certain type. $\mathcal{T}$
 $\mathcal{V} \subset \text{Diff}(M)$ such that
hyperbolic locally maxim
topologically conjugate*

Skew products

Note that skew products
dynamical systems.

Let Σ^2 be a space of bi

Let $\sigma : \Sigma^2 \mapsto \Sigma^2$ be a B

$$(\sigma(\omega))_k = \omega_{k+}$$

Definition 6. Let $\{f_\omega\}$
phisms of S^1 . A map f

from skew products to diffeomorphisms. A result of Gorodetski that can be stated as follows:

(Gorodetski, 2001). *Let $\mathcal{U}$ be a sufficient family in the space of skew products. Then there exists a C^1 -domain $\mathcal{V}$ such that any $f \in \mathcal{V}$ has a partially hyperbolic invariant set Δ , and $F|_{\Delta}$ is conjugate to a skew product from $\mathcal{U}$.*

These can be interpreted as random

infinite sequences of 0 and 1. The Bernoulli shift given by

$$T_k(\omega) = \omega_{k+1}, \quad \forall \omega \in \Sigma^2, k \in \mathbb{Z}.$$

$\{f_\omega\}_{\omega \in \Sigma^2}$ be a family of diffeomorphisms. A map $F: \Sigma^2 \times S^1 \mapsto \Sigma^2 \times S^1$ is called

which is a
Any such
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Since Cas
periodic p

Figure: F in
denotes fibre
Using dens
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and r_4 (cf
jectory is c
point p .
Using met

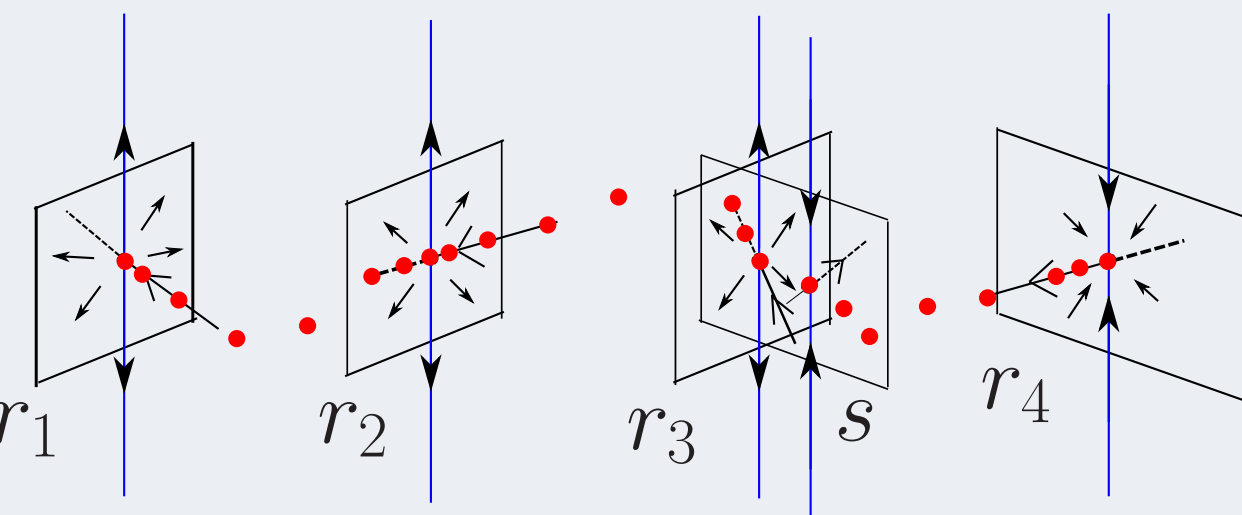
assigned to some mild skew product G .

F has the following property:

Any two periodic points with different indices are in the partially hyperbolic invariant set Δ .

Case 1 does not hold, for any two hyperbolic periodic points q_1 and q_2 lying on different fibres

$$W^u(q_1) \cap W^s(q_2) = \emptyset.$$



in Case 2: red denotes a pseudotrajectory, blue denotes fibres.

To study the density of hyperbolic periodic points with different indices, we construct a pseudotrajectory joining r_1 and r_4 (cf. the figure). Assume that this pseudotrajectory is orbitally close to an exact trajectory of some

methods from theory of skew products and sym

$$\xi \in N(\epsilon, O(p, f)).$$

Example of $f \in \text{WSP} \setminus \text{OSP}$: *the Plank*
(an Ω -stable diffeomorphism constructed by

Spaces of dynamical systems

Let M be a closed smooth Riemannian manifold. Let $H(M)$ denote the space of all homeomorphisms of M with C^0 -metric. By $\text{Diff}(M)$ denote the space of diffeomorphisms of M with C^1 -metric.

Definition 5. A set is called *generic* if it is a second category set (i.e. contains a countable union of open and dense sets) in the corresponding topology.

menevskaya map
ted in 1999).

an manifold. By
morphisms of M
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c.

c if it is a Baire
countable union
onding space.

a *mild skew product* iff

$$f(\omega, \phi) =$$

Let g_0 be a rotation of
Smale diffeomorphism w
 p_1 and p_2 , and $Df(p_2)$
the figure).

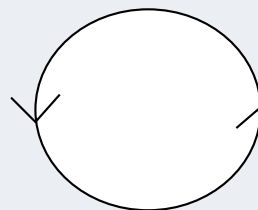


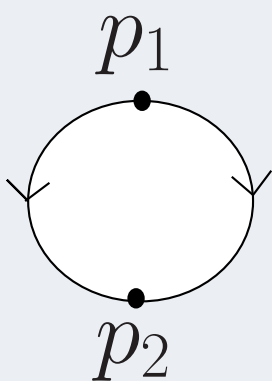
Figure: Diffeom

Let a be close to 1, b b
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products $\{f_\omega\}_{\omega \in \Sigma^2}$ such

- f_ω is δ -close to g_0 i
- f_ω is δ -close to g_1 i

$$= (\sigma(\omega), f_\omega(\phi)).$$

S^1 by angle b , g_1 be a Morse-
with two hyperbolic fixed points
 $= a < 1$, $Df(p_1) = 1/a$ (cf.



morphisms g_0 and g_1 .

be close to 0, and δ be a suffi-
cient $\mathcal{U}$ is a set of all mild skew
that

in $\text{Diff}(S^1)$ if $\omega_0 = 0$;

in $\text{Diff}(S^1)$ if $\omega_0 = 1$.

Using met
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which con
*Note that
tant to pr
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is the key*

Possible

- 1 **Conj**
two-d
- 2 One-s
- 3 Invers
- 4 Limit

Methods from theory of skew products and sym-
mics, we prove that

$$p \in W^u(r_1) \cap W^u(r_4)$$

contradicts to our conditions.

*in order to derive contradiction it is impor-
rove that the intermediate part of the pseu-
ory is far from the points r_1 and r_4 . In fact it
technical difficulty.*

Directions for further research.

Lecture: OSP is C^1 -generic for
dimensional manifolds.

sidered shadowing properties.

se shadowing properties.

shadowing.